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A Superimposed Pilot-Based Integrated Radar and Communication System

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ABSTRACT In this paper, a multiple-input multiple-output (MIMO) integrated radar and communication system based on orthogonal frequency division multiplexing (OFDM) signals is proposed. We consider a bistatic scenario. Exploiting superimposed pilots and orthogonal space-time block code (OSTBC) precoding, transmitted power can be effectively used in both information symbol sending and target sensing with little energy waste and performance degradation. The expectation maximization (EM) is employed to estimate unknown symbols and the channel jointly, which can take the place of matched filtering in traditional radar. In the EM algorithm, we propose to calculate the channel expectation by using Gaussian distribution of the symbol estimation, which can make full use of the OSTBC property of sensing spatial diversity with fast maximum likelihood decoding. The autocorrelation response, and the Doppler response of moving targets in slow time are verified by numerical experiments. Simulations also show superimposed pilot is suitable in the integrated radar and communication system, even when targets with large Doppler shift exist.

INDEX TERMS Integrated radar and communication, OFDM, MIMO, superimposed pilot, OSTBC.

I. INTRODUCTION

Recently, it has been gradually realized that wireless communication and radar sensing function should coexist and cooperate with each other in some applications, in order to allow hardware reuse, cost reduction and architecture unification. Attempts have been made to integrate wireless communication and target sensing within a single transceiver platform [1].

The multiple-input multiple-output (MIMO) radar is highly developing for recent years that attracts great research interests both in military and commercial applications [2]. MIMO radars can be divided into two categories according to the configuration of antennas: colocated MIMO radars, and statistical MIMO radars. The transmit and receiving antennas of a statistical MIMO radar are widely separated so that the radar has spatial diversity, and performance of target detection can be improved [3]. The flexibility of generating the probing signals offers a MIMO radar waveform diversity. It has the merits of higher resolution, higher sensitivity and better parameter identifiability.

Recently, a new kind MIMO radar is developed in [3]–[5] based on the orthogonal frequency division multiplexing (OFDM) signals. For communication applications, the OFDM waveform has been widely used in the digital television broadcasting, digital audio broadcasting, digital video broadcasting, wireless local area network, etc. This technique offers a number of merits, such as, flexible multi-carrier modulation, the availability of processing gain at the receiver, whitening of the signal spectrum, reduction of the power spectral density [6], and easy synchronization and equalization [7]. Furthermore, the orthogonality of the signals is not affected by the time delay from the transmitters, which provides the OFDM MIMO radar an inter-range-cell interference (IRCI) free property [3].

There is a vast literature dedicated to joint radar and communication systems [1]. For example, intelligent transportation systems require vehicles to have the capability of autonomously sensing driving environments and exchanging information data cooperatively. In [8], a mmWave joint vehicular communication and radar system at 60 GHz is built based on the IEEE 802.11ad mmWave WLAN standard, while automotive radar also leverages the same waveform. Fundamental limitations of designing such an integrated MIMO OFDM system are investigated in [7]. Suitable system parameters are obtained through an optimization method.

An important problem of the integrated system is the coexistence, cooperative operation of the radar and

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communication. There are several types of spectrum sharing techniques to avoid self-interference between the two sub- systems in spatial domain [9], [10], [11] and time domain [12]. More specifically, the spectrum sharing between a matrix completion based MIMO radar and a MIMO communication system is considered in [9]. The communication transmit covariance matrix is designed to minimize the effective interference power at the radar receiver, and maintain the capacity for the communication system at the same time. The radar clutter suppression is also taken into consideration therein. In [10], [11], sidelobe control of the transmit beamforming enables communication links using the same pulse radar spectrum. An integrated system using a single transceiver platform towards mmWave applications is presented in [12]. The dual functions are sequentially arranged in time domain, so that they can operate independently and also jointly.

In this paper, we propose an OFDM MIMO integrated radar and communication system exploiting superimposed pilots. It has the ability of delivering information symbols and sensing targets simultaneously, especially with little transmit energy waste or performance deterioration. The potential applications of the proposed joint system lie in high-power base stations or hot spots. The passive receivers can utilize the superimposed signals to detect targets flying at low altitudes, to monitor intruders, or to do the traffic surveillance.

Essentially, the channel estimation procedure using superimposed pilot signals in a communication system is similar with the sensing of a radar, which enlightens us on combining the two parts through coding into a dual-function system. In the proposed joint system, we apply the cyclic prefix(CP)-based OFDM waveform, and the orthogonal space-time block code (OSTBC) [3] to generate mutually orthogonal waveforms on different antennas. Superimposed pilots are used in the integrated system to estimate the channel and sense the targets simultaneously.

By exploiting superimposed pilots in the proposed system, we develop an expectation maximization (EM) algorithm to fulfil both functions by only transmitting one waveform. To the best of our knowledge, this kind of superimposed pilots have never been used in radar detection. It has great potential in future joint systems. In the proposed EM algorithm, the channel expectation is calculated by using Gaussian distribution of the symbol estimation, which can make full use of the OSTBC property of sensing full spatial diversity with fast maximum likelihood (ML) decoding. Comparing to the comb and block type pilots [13], superimposed pilots have advantages of both types, which are suitable for both frequency selective channels and fast fading channels.

The rest of this paper is organized as follows. Section II introduces the system model of the OFDM MIMO integrated radar and communication system. Section III elaborates on jointly estimating of channel and symbols by using EM algorithm. Supporting simulation results are presented in Section IV. The conclusions are drawn in Section V.

Notations: diag( ) denotes making a diagonal matrix with the elements of the vector. The notations $\|\cdot\|$, $\|\|_F^2$, tr$(\cdot)$ denote ℓ_2 norm, Frobenius norm, trace of the matrix, respectively. Notation $\otimes$ represents Kronecker product. The notation vec( ) denotes the vectorization operator of a matrix. The superscripts *, ^{T}, ^{H} denote the conjugation, transposition, conjugate transposition of the matrix, respectively. Notation $\mathbb{E}$ means the expectation operator. The notation Re( ) and Im( ) denote the real and imaginary part of a complex value. The notation:= means definition, and $\lceil\rceil$ denotes round-up.

II. SYSTEM MODEL

We consider a bistatic scenario with or without a line of sight. A bistatic OFDM statistical MIMO radar carries information symbols simultaneously as illustrated in Fig. 1. Some pilot symbols are also transmitted, which are known to both sides. In the receiver, the radar part is to sense the channel response caused by targets, while the communication part draws information transmitted through this channel, too. We assume the transmitter and the receiver have synchronizing pulses from a same time benchmark, such as the global positioning system.

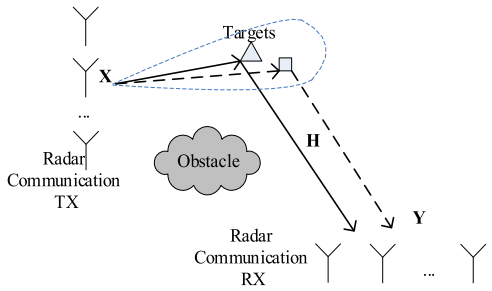


FIGURE 1. System model of joint radar communication.

A. OFDM MIMO SYSTEM

The transmitted signals are OFDM signals separately emitted through P antennas respectively. The radio frequency (RF) signal on the pth transmit antenna is:

$$x_p^{RF}(t) = \mathrm{Re}\left[e^{j2\pi f_c t} x_p(t)\right], \tag{1}$$

where f_c is the carrier frequency, and $x_p(t)$ is the baseband complex signal. Moreover, signal of the pth transmit antenna, $x_p(t)$, is given by:

$$x_p(t) = \frac{1}{K}\sum_{k=-K/2}^{K/2-1} X_p(k) e^{j2\pi k \Delta f t} g(t), \quad p = 0, \ldots, P-1, \tag{2}$$

where each OFDM symbol carries K subcarriers, expressed as $X_p(k)$ for the pth transmit antenna, which will be given in the subsection C. Notice that $x_p(t)$ is the inverse Fourier transform (IFT) of data symbols $X_p(k)$. The gate function $g(t)$ is a rectangular time window of length $T + T_{cp}$, i.e.,

$$g(t) = \begin{cases} 1 & t \in [-T_{cp}, T], \\ 0 & \text{others} \end{cases} \tag{3}$$

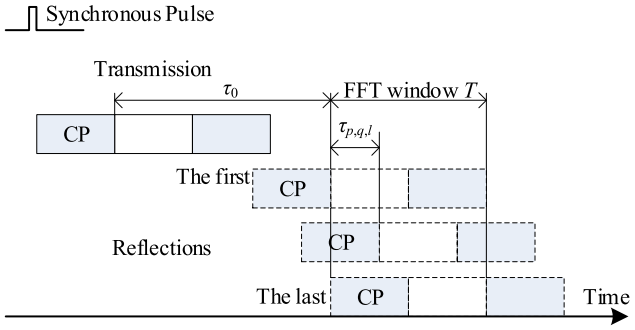


FIGURE 2. CP based OFDM waveform and the reflected signals.

where $T = 1/\Delta f$, and Δf is the frequency spacing of the subcarriers. The length of CP, T_{cp}, is used to add a guard signal before a data symbol in time domain [3], whose value equals the last segment of $x_p(t)$ within a duration, T_{cp}. The CP is used to efficiently estimate the channel response in an orthogonal and normalized manner by the receiver. Its length depends on the maximum surveillance distance of the radar [3]. 'MIMO' often means simultaneous transmission and reception using uncorrelated signals. If all P transmit antennas use the full K subcarriers, the orthogonality between antennas can be guaranteed through OSTBC method, which will be discussed in the subsection C.

Transmitted signals are propagated in free space, and are reflected by the targets. Reflected signals are intercepted by the receive antennas then. After down conversion from the RF and discretization in time domain, the baseband complex wave received from the qth receive antenna is:

$$y_q(n)=\sum_{p=0}^{P-1}\sum_{l=0}^{L-1}a_{p,q,l}x_p(nT_s-\tau_{p,q,l})e^{-j2\pi(f_{\mathrm{D}p,q,l}n+f_c\tau_{p,q,l})}+w_q(n),\quad q=0,\ldots Q-1, \tag{4}$$

where $n \in [0, K-1]$ is the discrete time indexing variable, $w_q(n)$ is the complex white Gaussian noise with zero-mean and the covariance σ^2, L is the number of the targets, P is the number of transmit antennas, Q is the number of the receiving antennas and $T_s = 1/(K\Delta f)$ is the sampling interval. The sampling begins at time τ_0, and $\tau_{p,q,l}$ is the time delay of the signal transmitted from the pth TX antenna reflected by the lth targets and received by the qth RX antenna relative to τ_0. The relationship of the time delays is illustrated in Fig. 2. Notation $a_{p,q,l}$ is the amplitude determined by radar cross section (RCS) of the lth targets, and $f_{\mathrm{D}p,q,l}$ is the corresponding Doppler shift frequency normalized by sampling interval T_s.

Thus, by inserting baseband complex wave of (2) into (4), we get

$$y_q(n)=\sum_{p=0}^{P-1}\sum_{l=0}^{L-1}g_{p,q,l}\frac{1}{K}\sum_{k=-K/2}^{K/2-1}X_p(k)e^{-j2\pi k\Delta f\tau_{p,q,l}}e^{j2\pi kn/N}+w_q(n), \tag{5}$$

where $g_{p,q,l} = a_{p,q,l}e^{-j2\pi f_c\tau_{p,q,l}}e^{-j2\pi f_{\mathrm{D}p,q,l}n_0}$, which is a constant independent of frequency index k. Assume the Doppler frequency is small enough to make the term $e^{-j2\pi f_{\mathrm{D}p,q,l}n}$ constant over the period of a symbol, so the term can be written as $e^{-j2\pi f_{\mathrm{D}p,q,l}n_0}$ where n_0 denotes the slow time of the symbol. Because the window length of $g(t)$ is $T+T_{cp}$, which is greater than the sampling window T, $g(t)$ is omitted in (5).

The fast Fourier transformation (FFT) is then carried out to transform $y_q(n)$ to its counterpart in frequency domain. Here, FFT window size is T. Its position is chosen to guarantee all the reflections from the concerned area can be covered in the FFT window, which means all the CPs of these signals must overlay the leading edge of the window, as illustrated in Fig. 2. The CP should be sufficient [3]. In practice, the length of CP and the location of FFT window are determined by the limit of radar range. This condition keeps the orthogonality of the signals in frequency domain [5]. Note that $x_p(nT_s-\tau_{p,q,l})$ in (4) is actually cyclic right shift version of $x_p(t)$ in (2) because of CP. Thus, after FFT, the signal in frequency domain, considering the reflected signals of a train of I symbols, is given by:

$$Y_{q,i}(k)=\sum_{p=0}^{P-1}X_{p,i}(k)\sum_{l=0}^{L-1}g_{p,q,l}e^{-j2\pi k\Delta f\tau_{p,q,l}}+W_{q,i}(k), \tag{6}$$

where i represents the index in a block of I symbols. $W_{q,i}(k)$ is the Fourier transform of $w_q(n)$ of the ith symbol, which is still complex Gaussian noise with zero-mean and the variance σ^2, since the FFT is a unitary transformation. We also assume that the channel response is constant within the period of I consecutive symbols, which is called the coherent processing interval (CPI). More detailed derivation of (6) is presented in [14]. The received signal model in (6) after FFT can also be expressed in a matrix form:

$$\mathbf{Y}(k)=\mathbf{H}_{\mathrm{f}}(k)\mathbf{X}(k)+\mathbf{W}(k), \tag{7}$$

where $\mathbf{X}(k) \in \mathbb{C}^{P\times I}$ is the transmit signal in frequency domain, and $\mathbf{Y}(k) \in \mathbb{C}^{Q\times I}$. $\mathbf{W}(k) \in \mathbb{C}^{Q\times I}$ is complex Gaussian noise with zero-mean and the variance σ^2. $\mathbf{H}_{\mathrm{f}}(k) \in \mathbb{C}^{Q\times P}$ is the channel frequency response of the kth carrier, which is defined as (8), as shown at the bottom of this page.

$$\mathbf{H}_{\mathrm{f}}(k)=\sum_{l=0}^{L-1}\begin{bmatrix} g_{0,0,l}e^{-j2\pi k\Delta f\tau_{0,0,l}} & \cdots & g_{P-1,0,l}e^{-j2\pi k\Delta f\tau_{P-1,0,l}} \\ \cdots & \cdots & \cdots \\ g_{0,Q-1,l}e^{-j2\pi k\Delta f\tau_{0,Q-1,l}} & \cdots & g_{P-1,Q-1,l}e^{-j2\pi k\Delta f\tau_{P-1,Q-1,l}} \end{bmatrix} \tag{8}$$

B. TARGET RESPONSE FOR RADAR

The frequency orthogonality of OFDM makes it convenient in expressing the received signals as the product of the transmitted signals and the channel response in frequency domain, as demonstrated in (7). The channel response, $\mathbf{H}_{\mathrm{f}}(k)$, in (8) caused by targets is expressed in frequency domain. And $\mathbf{H}_{\mathrm{f}}(k)$ can also be represented by the Fourier transform of the channel impulse response in time domain on N range cells, $\mathbf{H}_{\mathrm{t}}(n) \in \mathbb{C}^{Q\times P}$, $n = 0, \ldots, N-1$, i.e.,

$$\mathbf{H}_{\mathrm{f}}(k) = \sum_{n=0}^{N-1} \mathbf{H}_{\mathrm{t}}(n) e^{-j2\pi nk/K}. \tag{9}$$

The above FFT of the channel response can also be expressed in a matrix form as:

$$\mathbf{H}_{\mathrm{f}}(k) = \mathbf{H}(\mathbf{f}(k) \otimes \mathbf{I}_P), \tag{10}$$

where $\mathbf{H} := [\mathbf{H}_{\mathrm{t}}(0), \ldots, \mathbf{H}_{\mathrm{t}}(N-1)] \in \mathbb{C}^{Q\times PN}$ is the total channel response in time domain, and $\mathbf{f}(k) \in \mathbb{C}^{N\times 1}$ is the Fourier transform vector for the kth frequency defined as $\mathbf{f}(k) := [1, e^{-j2\pi k/K}, \ldots, e^{-j2\pi(N-1)k/K}]$.

The 'channel response' is usually referred to a term in communication. In fact, channel response in time domain is actually one dimensional range profile for radar. If the targets lies in different range cells, they can be identified through estimating channel impulse response $\mathbf{H}$. We will show this by calculating the channel response in time domain. Let $h_{p,q}(n)$ denote the (p, q)th entry of $\mathbf{H}_{\mathrm{t}}(n)$, and it is the inverse Fourier transform of $\mathbf{H}_{\mathrm{f}}(k)$:

$$h_{p,q}(n) = \frac{1}{K} \sum_{k=0}^{K-1} \sum_{l=0}^{L-1} g_{p,q,l} e^{-j2\pi k \Delta f \tau_{p,q,l}} e^{j2\pi nk/K} \tag{11}$$

By using the Dirichlet kernel [15], $h_{p,q}(n)$ can be expresses as:

$$h_{p,q}(n) = \frac{1}{K} \sum_{l=0}^{L-1} g_{p,q,l} \frac{\sin\left[\pi\left(\Delta f \tau_{p,q,l} - n/K\right)K\right]}{\sin\left[\pi\left(\Delta f \tau_{p,q,l} - n/K\right)\right]} \times e^{-j\pi\left(\Delta f \tau_{p,q,l} - n/K\right)(K-1)}. \tag{12}$$

Notice that the peak for every target l appears at time $n = round(K\Delta f \tau_{p,q,l})$ in the (p, q)th range profile because of the discrete sampling interval. An example of a range profile is illustrated in Fig. 3, with only one target existing in the observing window. The red dashed lined with circle marks represents a target whose delay is exactly on the grid of sampling, while the green dotted line is that of $T_s/4$ offset and the blue dash-dotted line is for $T_s/2$ offset. As one can observe from (12), the channel response of off grid targets shows non-zero sidelobes, while on grid targets result in ideally zero sidelobes in range which is called IRCI free in [3]. Off grid is more general in practical applications. We will not restrict to the on grid condition in this paper.

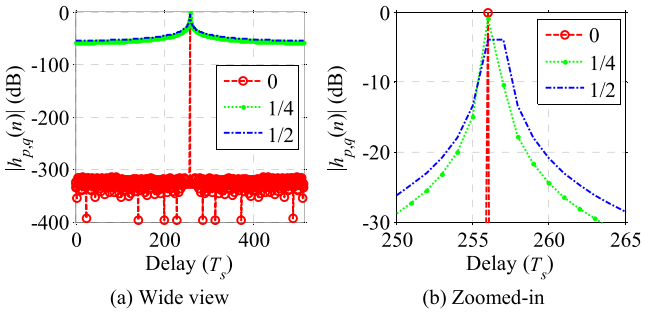


FIGURE 3. An example of the channel response with one target. (a) Wide view. (b) Zoomed-in.

A theoretical one dimensional range profile of an OFDM based radar is given in (12). Next the problem becomes how to estimate the channel response $\mathbf{H}$ from a reflected signals with some unknown information symbols.

C. SUPERIMPOSED OSTBC CODING

For radar sensing, the transmit symbols $\mathbf{X}(k)$ should be known to the receiver, while the communication delivers unknown symbols. Of course, communication also transmits pilot symbols known to both sides to identify the channel response. In this sense, two distinct systems have something in common.

OFDM technique exploits various interpolation methods to estimate the channel response of subcarriers. Different types of pilot structures are considered in terms of the arrangement of pilots, such as block type, comb type [13]. Block type pilots occupy all the subcarriers of an OFDM symbol, and are transmitted periodically along the time axis. A time-domain interpolation is carried out to estimate the channel, so it is suitable for frequency selective channel, not for fast fading channels. On the contrary, comb type shows the opposite characteristic.

Generally speaking, block type is not suitable for large Doppler targets, while comb type is not suitable for non-sparse channels in time domain. We propose superimposed pilots to satisfy both the conditions in our integrated radar and communication system. This type of pilot is studied in channel estimation of communication system as in [16], [17]. It can transmit pilots along with the information symbols simultaneously to suit both channel types. Moreover, it can obtain optimal performance only by adjusting the ratio of pilot and information power. This is more flexible than the other types of pilot structures.

The transmitted signal for each carrier k can be expressed as:

$$\mathbf{X}(k) = \mathbf{S}_{\mathrm{c}}(k)\mathbf{U} + \mathbf{P}_{\mathrm{c}}(k)\mathbf{V}, \tag{13}$$

where $\mathbf{S}_{\mathrm{c}}(k) \in \mathbb{C}^{P\times(I-R)}$ is the information symbol matrix coded by OSTBC and $\mathbf{U} \in \mathbb{C}^{(I-R)\times I}$, $\mathbf{V} \in \mathbb{C}^{R\times I}$ are the superimposed precoding matrixes. The matrix of pilot symbols, $\mathbf{P}_{\mathrm{c}}(k) \in \mathbb{C}^{P\times R}$, is of size $P \times R$, $(R \geq P)$, and $R = P$ is the minimum number required by a MIMO system with P transmit antennas to obtain the channel response [17]. The design of pilot symbols will be discussed in the next section.

The information-symbol matrix coded by OSTBC is given by [18]

$$\mathbf{S}_{\mathrm{c}}(k) = \mathbf{A}(\mathrm{Re}(\mathbf{S}(k)) \otimes \mathbf{I}_{N_{\mathrm{c}}}) + j\mathbf{B}(\mathrm{Im}(\mathbf{S}(k)) \otimes \mathbf{I}_{N_{\mathrm{c}}}), \tag{14}$$

where N_{s} is the number of the original symbols, N_{c} is the length of the codeword of OSTBC, and $\mathbf{A} := [\mathbf{A}_0, \ldots, \mathbf{A}_{N_{\mathrm{s}}-1}] \in \mathbb{R}^{P\times N_{\mathrm{c}}N_{\mathrm{s}}}$, $\mathbf{B} := [\mathbf{B}_0, \ldots, \mathbf{B}_{N_{\mathrm{s}}-1}] \in \mathbb{R}^{P\times N_{\mathrm{c}}N_{\mathrm{s}}}$. The information symbol matrix in (14) is composed of B consecutive OSTBC codewords, i.e., $\mathbf{S}(k) \in \mathbb{C}^{N_{\mathrm{s}}\times B}$, where $B = (I - R)/N_{\mathrm{c}}$. The elements of $\mathbf{S}(k)$ are modeled as zero mean i.i.d. random variables with variance σ_s^2.

A set of OSTBC coding matrixes $\mathbf{A}_\xi, \mathbf{B}_\xi \in \mathbb{R}^{P\times N_{\mathrm{c}}}$, $\xi = 0, \ldots, N_{\mathrm{s}} - 1$, is called an amicable orthogonal design. An amicable orthogonal design has some good properties to guarantee decoupled ML detection easy to derive [18]. As an example, an OSTBC coded symbol for antennas $P = N_{\mathrm{s}} = N_{\mathrm{c}} = 2$ and $B = 1$ is demonstrated as:

$$\mathbf{S}(k) = \begin{bmatrix} s_0(k) \\ s_1(k) \end{bmatrix}, \quad \mathbf{S}_{\mathrm{c}}(k) = \begin{bmatrix} s_0(k) & s_1^*(k) \\ s_1(k) & -s_0^*(k) \end{bmatrix}, \tag{15}$$

which is also known as Alamouti code. The coding matrixes $\mathbf{A}_l, \mathbf{B}_l$ in the above example are given by:

$$\begin{aligned} \mathbf{A}_0 &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{A}_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \mathbf{B}_0 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B}_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \end{aligned} \tag{16}$$

We only show a coding method for codes $P = 2$ in the example. In fact, given the number of antennas P, the optimal numbers of N_{c} and N_{s} are determined by P [19]. The relationship between the three variables is given by [19]:

$$\frac{N_{\mathrm{s}}}{N_{\mathrm{c}}} = \frac{\lceil P/2 \rceil + 1}{2\lceil P/2 \rceil}. \tag{17}$$

Some construction examples for greater P, up to 7, are also given in [20], and a table of the relationship is listed therein. Note that as the number of antennas P grows, more symbols N_s will be contained in a block.

The superimposed precoding matrixes $\mathbf{U}$ and $\mathbf{V}$ are get from an arbitrary unitary matrix [16]:

$$\mathbf{O} = \begin{bmatrix} \mathbf{U} \\ \mathbf{V} \end{bmatrix}, \quad \mathbf{O}\,\mathbf{O}^H = \mathbf{I}_I, \tag{18}$$

which implies $\mathbf{U}\mathbf{U}^H = \mathbf{I}_{(I-R)}$, $\mathbf{V}\mathbf{V}^H = \mathbf{I}_R$ and $\mathbf{U}\mathbf{V}^H = \mathbf{0}_{(I-R)\times R}$. Linearly precoding the information symbols and the pilot sequences using $\mathbf{U}$ and $\mathbf{V}$ can make them orthogonal to each other. Therefore, $\mathbf{U}$ and $\mathbf{V}$ can be used to get rid of information symbols when estimating channel response, or to null pilots when detecting information symbols.

To our best knowledge, [21] is the first to introduce STBC to the MIMO radar to reduce waveform cross correlation. OSTBC is used in [3] for the sake of structured orthogonality. Here, we also adopt OSTBC combined with superimposed orthogonal coding $\mathbf{U}$ and $\mathbf{V}$ to generate uncorrelated signals for radar. The orthogonality property of $\mathbf{U}$, $\mathbf{V}$ and OSTBC will play a role in Section III and Appendix D.

D. THE OVERALL SYSTEM MODEL

By substituting (13) into (7), and stacking signals for all the K sub-carriers, we get a very compact system model:

$$\mathbf{Y} = \mathbf{H}\mathbf{X} + \mathbf{W}, \tag{19}$$

where

$$\mathbf{Y} := [\mathbf{Y}(0), \ldots, \mathbf{Y}(K-1)] \in \mathbb{C}^{Q\times IK}, \tag{20}$$

$$\mathbf{W} := [\mathbf{W}(0), \ldots, \mathbf{W}(K-1)] \in \mathbb{C}^{Q\times IK}, \tag{21}$$

and $\mathbf{H}$ is defined below (10). This model integrates the channel responses for all the subcarriers, and $\mathbf{H}$ is the target response in time domain, which makes is easy to get the radar range profile. The overall transmit signal $\mathbf{X} \in \mathbb{C}^{PN\times IK}$ is given by:

$$\begin{aligned} \mathbf{X} &:= [(\mathbf{f}(0) \otimes \mathbf{I}_P)\mathbf{X}(0), \ldots, (\mathbf{f}(K-1) \otimes \mathbf{I}_P)\mathbf{X}(K-1)], \\ &= \mathbf{S}_{\mathrm{U}} + \mathbf{P}_{\mathrm{V}} \end{aligned} \tag{22}$$

where $\mathbf{S}_{\mathrm{U}} \in \mathbb{C}^{PN\times IK}$, and $\mathbf{P}_{\mathrm{V}} \in \mathbb{C}^{PN\times IK}$ is defined as follows:

$$\mathbf{S}_{\mathrm{U}} = [(\mathbf{f}(0) \otimes \mathbf{I}_P)\mathbf{S}_{\mathrm{c}}(0)\mathbf{U}, \ldots, (\mathbf{f}(K-1) \otimes \mathbf{I}_P)\mathbf{S}_{\mathrm{c}}(K-1)\mathbf{U}], \tag{23}$$

$$\mathbf{P}_{\mathrm{V}} = [(\mathbf{f}(0) \otimes \mathbf{I}_P)\mathbf{P}_{\mathrm{c}}(0)\mathbf{V}, \ldots, (\mathbf{f}(K-1) \otimes \mathbf{I}_P)\mathbf{P}_{\mathrm{c}}(K-1)\mathbf{V}]. \tag{24}$$

III. JOINTLY SENSING AND SYMBOL ESTIMATION

In conventional radar, a matched filter is employed to obtain enough gain to detect targets. In the bistatic integrated system, however, not all the signals transmitted are known beforehand to the receiver. We propose a jointly sensing and unknown symbol estimation method based on EM to attain a result comparable to the matched filtering. At first, superimposed pilots are used to estimate the channel roughly. After the information symbols are detected, the pilots together with the estimated information symbols are used to sense targets for radar in an iterative way.

A. INFORMATION AND PILOT SEPARATION

For the superimposed system, we should separate the pilots and the information symbols first. Because of the property of superimposed precoding matrixes, post-multiplying (19) with $\mathbf{I}_K \otimes \mathbf{V}^H$ may yield:

$$\mathbf{Y}_{\mathrm{p}} := \mathbf{Y}(\mathbf{I}_K \otimes \mathbf{V}^H) = \mathbf{H}\mathbf{P} + \mathbf{W}(\mathbf{I}_K \otimes \mathbf{V}^H), \tag{25}$$

where

$$\begin{aligned} \mathbf{P} = [(\mathbf{f}(0) \otimes \mathbf{I}_P)\mathbf{P}_{\mathrm{c}}(0), \ldots, (\mathbf{f}(K-1) \otimes \mathbf{I}_P)\mathbf{P}_{\mathrm{c}}(K-1)] \\ \times \in \mathbb{C}^{PN\times RK}. \end{aligned} \tag{26}$$

The equation holds because of the orthogonal property of superimposed precoding matrixes $\mathbf{U}$ and $\mathbf{V}$, which ensures that pilot and data symbols are separable. From (25) one can observe that the unknown information symbols are eliminated from overall receiving signals by the orthogonal precoding matrixes, so that this measurement equation can be used to estimate the channel response $\mathbf{H}$.

In the same way, information symbols can also be separated from mixed receiving signals by post-multiplying (19) by $\mathbf{I}_K \otimes \mathbf{U}^H$ to nullify the impact of pilots as follows:

$$\mathbf{Y}_\mathrm{d} := \mathbf{Y}(\mathbf{I}_K \otimes \mathbf{U}^H) = \mathbf{HS} + \mathbf{W}(\mathbf{I}_K \otimes \mathbf{U}^H), \tag{27}$$

where

$$\mathbf{S} = [(\mathbf{f}(0) \otimes \mathbf{I}_P)\mathbf{S}_\mathrm{c}(0), \ldots, (\mathbf{f}(K-1) \otimes \mathbf{I}_P)\mathbf{S}_\mathrm{c}(K-1)] \times \in \mathbb{C}^{PN\times(I-R)K}. \tag{28}$$

Combine (25) and (27) to get a new version of signal model with information and pilots being separated:

$$[\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}] = \mathbf{H}[\mathbf{P}, \mathbf{S}] + [\mathbf{W}(\mathbf{I}_K \otimes \mathbf{V}^H), \mathbf{W}(\mathbf{I}_K \otimes \mathbf{U}^H)], \tag{29}$$

where entities in $[\mathbf{W}(\mathbf{I}_K \otimes \mathbf{V}^H), \mathbf{W}(\mathbf{I}_K \otimes \mathbf{U}^H)]$ are complex Gaussian variables with zero-mean and variance σ^2 since the precoding matrixes are unitary.

B. SYMBOL DETECTION

Because (27) is formed by stacking all the subcarriers together, the received signals of (27) can be divided into expressions of separated sub-carriers in order to estimate information symbols more conveniently:

$$\mathbf{Y}(k)\mathbf{U}^H = \mathbf{H}_\mathrm{f}(k)[\mathbf{A}(\mathrm{Re}(\mathbf{S}(k)) \otimes \mathbf{I}_{N_\mathrm{c}}) + j\mathbf{B}(\mathrm{Im}(\mathbf{S}(k)) \otimes \mathbf{I}_{N_\mathrm{c}})] + \mathbf{W}(k)\mathbf{U}^H. \tag{30}$$

Because $\mathbf{S}(k)$ is composed of B consecutive OSTBC codewords, dividing $\mathbf{Y}(k)\mathbf{U}^H$ into B block sub-matrixes of the same size, vectorizing all the sub-matrixes and then stacking them, we have an effective model for data detection:

$$\mathbf{Y}_\mathrm{U}(k) = \mathbf{G}_\mathrm{A}(k)\mathrm{Re}(\mathbf{S}(k)) + j\mathbf{G}_\mathrm{B}(k)\mathrm{Im}(\mathbf{S}(k)) + \mathbf{W}_\mathrm{U}(k), \tag{31}$$

where $\mathbf{W}_\mathrm{U}(k)$ is noise term (see more in Appendix A), $\mathbf{Y}_\mathrm{U}(k) \in \mathbb{C}^{QN_\mathrm{c}\times B}$ is a reshaped version of the received signal with superimposed pilots already decoupled from the original one. It can be expressed as:

$$\mathbf{Y}_\mathrm{U}(k) := \left\lfloor \mathrm{vec}(\mathbf{Y}(k)\mathbf{U}_0^H), \ldots, \mathrm{vec}(\mathbf{Y}(k)\mathbf{U}_{B-1}^H) \right\rfloor, \tag{32}$$

where

$$\mathbf{U} = \begin{bmatrix} \mathbf{U}_0 \\ \vdots \\ \mathbf{U}_{B-1} \end{bmatrix}, \tag{33}$$

and $\mathbf{U}_b \in \mathbb{C}^{N_\mathrm{c}\times I}$, $b = 0, \ldots, B-1$ is the block sub-matrix of $\mathbf{U}$ composed of rows from number (bN_c) to $((b+1)N_\mathrm{c}-1)$. The matrixes $\mathbf{G}_\mathrm{A}(k)$ and $\mathbf{G}_\mathrm{B}(k)$ in (31) are given by:

$$\mathbf{G}_\mathrm{A}(k) := [\mathrm{vec}(\mathbf{H}_\mathrm{f}(k)\mathbf{A}_0), \ldots, \mathrm{vec}(\mathbf{H}_\mathrm{f}(k)\mathbf{A}_{N_\mathrm{s}-1})] \in \mathbb{C}^{QN_\mathrm{c}\times N_\mathrm{s}},$$
$$\mathbf{G}_\mathrm{B}(k) := [j\mathrm{vec}(\mathbf{H}_\mathrm{f}(k)\mathbf{B}_0), \ldots, j\mathrm{vec}(\mathbf{H}_\mathrm{f}(k)\mathbf{B}_{N_\mathrm{s}-1})]. \tag{34}$$

Pre-multiplying both sides of (31) by $\mathbf{G}_\mathrm{A}^H(k)$ and $\mathbf{G}_\mathrm{B}^H(k)$ respectively, one can get:

$$\mathrm{Re}\left(\mathbf{G}_\mathrm{A}^H(k)\mathbf{Y}_\mathrm{U}(k)\right) = \|\mathbf{H}_\mathrm{f}(k)\|_F^2 \mathrm{Re}(\mathbf{S}(k)) + \mathrm{Re}\left(\mathbf{G}_\mathrm{A}^H(k)\mathbf{W}_\mathrm{U}(k)\right),$$

$$\mathrm{Re}\left(\mathbf{G}_\mathrm{B}^H(k)\mathbf{Y}_\mathrm{U}(k)\right) = \|\mathbf{H}_\mathrm{f}(k)\|_F^2 \mathrm{Im}(\mathbf{S}(k)) + \mathrm{Re}\left(\mathbf{G}_\mathrm{B}^H(k)\mathbf{W}_\mathrm{U}(k)\right). \tag{35}$$

The equation holds because $\mathbf{G}_\mathrm{A}(k)$ and $\mathbf{G}_\mathrm{B}(k)$ satisfy the following relationship [17], [18]:

$$\mathrm{Re}\left(\mathbf{G}_\mathrm{A}^H(k)\mathbf{G}_\mathrm{B}(k)\right) = \mathrm{Re}\left(\mathbf{G}_\mathrm{B}^H(k)\mathbf{G}_\mathrm{A}(k)\right) = \mathbf{0},$$
$$\mathrm{Re}\left(\mathbf{G}_\mathrm{A}^H(k)\mathbf{G}_\mathrm{A}(k)\right) = \mathrm{Re}\left(\mathbf{G}_\mathrm{B}^H(k)\mathbf{G}_\mathrm{B}(k)\right) = \|\mathbf{H}_\mathrm{f}(k)\|_F^2\mathbf{I}_{N_\mathrm{s}}, \tag{36}$$

where the orthogonal property of OSTBC is utilized.

Provided that $\hat{\mathbf{H}}_\mathrm{f}(k)$ is the estimation of the channel response, the estimation of $\mathbf{S}(k)$ from (35) using ML approach can be expressed as:

$$\hat{\mathbf{S}}(k) = \left(\mathrm{Re}\left(\hat{\mathbf{G}}_\mathrm{A}^H(k)\mathbf{Y}_\mathrm{U}(k)\right) + j\mathrm{Re}\left(\hat{\mathbf{G}}_\mathrm{B}^H(k)\mathbf{Y}_\mathrm{U}(k)\right)\right) / \|\hat{\mathbf{H}}_\mathrm{f}(k)\|_F^2, \tag{37}$$

where $\hat{\mathbf{G}}_\mathrm{A}$ and $\hat{\mathbf{G}}_\mathrm{B}$ are defined by (34) with $\mathbf{H}_\mathrm{f}(k)$ being replaced by $\hat{\mathbf{H}}_\mathrm{f}(k)$. Notice that in the estimation formulation of $\mathbf{S}(k)$, noise $\mathbf{W}_\mathrm{U}(k)$ exists in $\mathbf{Y}_\mathrm{U}(k)$ as shown in (31). Thus, estimation error arises because of the noise. The true value of $\mathbf{S}(k)$ can be modeled as its estimation plus an estimation error:

$$\mathbf{S}(k) = \hat{\mathbf{S}}(k) - \mathbf{E}_\mathrm{s}(k), \tag{38}$$

where $\mathbf{E}_\mathrm{s}(k)$ is the symbol estimation error. The statistical property of this estimation error is given in Appendix B.

C. INITIALIZATION STAGE OF EM ALGORITHM

The EM algorithm is applied in estimating procedure of the channel and information simultaneously. EM algorithm is an iterative process that might converge to a local optimization. Hence EM initialization should be carried out carefully to guarantee converging to a global optimization point [22].

The channel response can be estimated using the pilot sequence of (25). The initial value of an EM algorithm is obtained from ML estimate of the channel matrix as [23]:

$$\hat{\mathbf{H}}^{(0)} = \mathbf{Y}_\mathrm{p}\mathbf{P}^H(\mathbf{P}\mathbf{P}^H)^{-1}. \tag{39}$$

From (39), one can see that the channel is estimated only by using the pilot symbols. If the data symbols can also be used to estimate the channel, the performance will be improved which is meaningful especially for radar detection.

The ML estimation of $\mathbf{H}$ using both the training and the data symbols is given by:

$$\hat{\mathbf{H}}_{ML} = \arg\max_{\mathbf{H}} \log p([\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}]|\mathbf{H}). \tag{40}$$

It is intractable to get a close-form solution to this ML problem of (40) [24]. Alternatively, an EM algorithm is proposed to solve it in an iterative way using the complete data [25].

D. EM ALGORITHM WITH GAUSSIAN PRIOR OF SYMBOL ESTIMATION

Here both the pilot and information symbols are used for channel estimation. The set $\{\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}, \mathbf{S}\}$ denotes the complete data space. EM algorithm comprises of two steps, an expectation evaluation and a maximization.

1) EXPECTATION STEP

Conditional expectation of log-likelihood function (LLF) of the complete data is calculated as follows, similar to [24]:

$$\begin{aligned}\mathcal{L}(\mathbf{H}|\hat{\mathbf{H}}^{(\rho-1)}) &:= \mathbb{E}_{p(\mathbf{S}|\mathbf{Y}_\mathrm{p},\mathbf{Y}_\mathrm{d},\hat{\mathbf{H}}^{(\rho-1)})} \log p(\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}, \mathbf{S}|\mathbf{H}) \\ &= \mathbb{E}_{p(\mathbf{S}|\mathbf{Y}_\mathrm{p},\mathbf{Y}_\mathrm{d},\hat{\mathbf{H}}^{(\rho-1)})} \log p(\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}|\mathbf{S}, \mathbf{H}) p(\mathbf{S}|\mathbf{H})\end{aligned} \tag{41}$$

where $p(\mathbf{S}|\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}, \hat{\mathbf{H}}^{(\rho-1)})$ is conditional density of $\mathbf{S}$ given $\mathbf{Y}_\mathrm{p}$, $\mathbf{Y}_\mathrm{d}$ and the old estimation of channel response $\hat{\mathbf{H}}^{(\rho-1)}$.

In (41), a conditional expectation should be calculated to get the maximum point. If we should suppose data symbols $\mathbf{S}$ are chosen from a discrete constellation, it turns out that the computational complexity of the expectation grows exponentially with the number of symbols. Alternatively, in [23], a zero-mean Gaussian prior distribution of $\mathbf{S}$ is proposed to evaluate the expectation.

2) MAXIMIZATION STEP

The EM algorithm updates the channel estimation at the ρth iteration, $\hat{\mathbf{H}}^{(\rho)}$, using the old estimation $\hat{\mathbf{H}}^{(\rho-1)}$ as follows instead of (40):

$$\hat{\mathbf{H}}^{(\rho)} = \arg\max_{\mathbf{H}} \mathcal{L}(\mathbf{H}|\hat{\mathbf{H}}^{(\rho-1)}). \tag{42}$$

The maximum point of the equation (42) is calculated by taking its complex gradient with respect to $\mathbf{H}$, and setting the gradient to zero and solving the equation (see more details in Appendix C). One can obtain the maximum value at the ρth iteration as follows:

$$\hat{\mathbf{H}}^{(\rho)} = \left(\mathbf{Y}_\mathrm{p}\mathbf{P}^H + \mathbf{Y}_\mathrm{d}\mathbf{M}_\mathrm{d}^H\right)\left(\mathbf{P}\mathbf{P}^H + \Sigma_\mathrm{d}\right)^{-1}, \tag{43}$$

where

$$\mathbf{M}_\mathrm{d} := \mathbb{E}\left(\mathbf{S}|\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}, \hat{\mathbf{H}}^{(\rho-1)}\right), \tag{44}$$

$$\mathbf{\Sigma}_\mathrm{d} := \mathbb{E}\left(\mathbf{S}\mathbf{S}^H|\mathbf{Y}_\mathrm{p}, \mathbf{Y}_\mathrm{d}, \hat{\mathbf{H}}^{(\rho-1)}\right). \tag{45}$$

Derivation of equation (43) involves some matrix manipulation and linearity of expectation operator.

The information symbols are OSTBC coded, so one can make full use of the properties of OSTBC to improve the convergent rate of the EM algorithm. Here, we propose to exploit a Gaussian prior distribution with its mean value equal to the transmit symbol estimation instead of the non-information Gaussian prior as in [23].

To take full advantage structure of OSTBC, the information symbol's ML estimation is used to evaluate LLF. The prior distribution of OSTBC coded data symbols $\mathbf{S}$ are assumed to be Gaussian with non-zero mean, in order to compute $\mathbf{M}_\mathrm{d}$ and $\mathbf{\Sigma}_\mathrm{d}$ in (43) (derivations presented in Appendix D). We get the approximated value for the ρth iteration as follows:

$$\mathbf{M}_\mathrm{d} = \hat{\mathbf{S}}^{(\rho)}, \tag{46}$$

$$\mathbf{\Sigma}_\mathrm{d} = \sum_{k=0}^{K-1} ||\hat{\mathbf{S}}^{(\rho)}(k)||_F^2 \mathbf{I}_P + KN_\mathrm{s}\sigma^2\mathbf{I}_P, \tag{47}$$

where $\hat{\mathbf{S}}^{(\rho)}$ and $\hat{\mathbf{S}}^{(\rho)}(k)$ are the estimation of $\mathbf{S}$ and $\mathbf{S}(k)$ at ρth iteration defined in Appendix D and (37) respectively. Substituting (46) and (47) in (43), the channel estimation for the ρth iteration is finally obtained as:

$$\begin{aligned}\hat{\mathbf{H}}^{(\rho)} = &\left(\mathbf{Y}_\mathrm{p}\mathbf{P}^H + \mathbf{Y}_\mathrm{d}\hat{\mathbf{S}}^{(\rho)H}\right) \\ &\times \left(\mathbf{P}\mathbf{P}^H + \sum_{k=0}^{K-1} ||\hat{\mathbf{S}}^{(\rho)}(k)||_F^2 \mathbf{I}_P + KN_\mathrm{s}\sigma^2\mathbf{I}_P\right)^{-1}.\end{aligned} \tag{48}$$

The overall steps of the proposed EM algorithm are summarized in Table 1.

TABLE 1. Jointly sensing and symbol estimation based on EM algorithm.

1. Set ρ=0, and get an initial value $\hat{\mathbf{H}}^{(0)}$ using (39) from the pilots only.
2. ρ= ρ+1.
3. Estimate the information symbols $\hat{\mathbf{S}}^{(\rho)}(k)$ by (37).
4. Calculate the channel response $\hat{\mathbf{H}}^{(\rho)}$ from (48).
5. Repeat step 2 through 4 until a stop criterion meets.

Notice that once we get the channel estimation $\hat{\mathbf{H}}$ using the EM algorithm, the one dimensional range profile for radar is get at the same time. According to the definition of the channel model $\mathbf{H}$ in (10), one only needs to rearrange the matrix $\hat{\mathbf{H}}$ to obtain the radar range profile.

E. PILOT DESIGN

The pilot symbols should be carefully designed to improve channel estimation performance as shown in some spatially correlated MIMO systems [16], [17]. In the integrated system, an important task is to sense the targets. Notice that linear frequency modulation (LFM) waveform used in radar have good properties in the pulse compression and ambiguity, and it is a good candidate for distinguishing point targets and imaging. A discrete version of the LFM signal, or so called Zadoff-Chu (ZC) sequence [26], is applied here:

$$\phi(n) = e^{-j\pi n(n+K \bmod 2)/K}. \tag{49}$$

The root index of the ZC sequence uses here happens to be 1, because the bandwidth of the OFDM signal is $K\Delta f$, and the duration of the OFDM is $T = 1/\Delta f$. The discrete Fourier transform (DFT) of the sequence is expressed as [27]:

$$\Phi(k) = \phi^*(k)\Phi(0). \tag{50}$$

Hence, ZC sequence has constant modulus in both time and frequency domain.

Next, in order to generate orthogonal waveforms on different transmit antennas, the same OSTBC method is employed in the pilot design. Same with (14), the OSTBC coded pilot is given by:

$$\mathbf{P}_{\mathrm{c}}(k) = \mathbf{A}(\mathrm{Re}(\mathbf{p}(k)) \otimes \mathbf{I}_{N_{\mathrm{c}}}) + j\mathbf{B}(\mathrm{Im}(\mathbf{p}(k)) \otimes \mathbf{I}_{N_{\mathrm{c}}}), \quad (51)$$

where the columns of the pilot $\mathbf{P}_{\mathrm{c}}(k)$ becomes $R = N_{\mathrm{c}}$, and $\mathbf{p}(k)$ is defined as:

$$\mathbf{p}(k) = [\Phi(k), \ldots, \Phi(k)]^T \in \mathbb{C}^{N_{\mathrm{s}} \times 1}. \quad (52)$$

Notice that the elements of the vector $\mathbf{p}(k)$ are all the same. That will not affect the orthogonality of the pilots on different antennas because of the OSTBC of (51) [21]. We will show in the numerical simulation OSTBC coded ZC sequence can also slightly reduce the peak-to-average power ratio (PAPR) of the original information symbol.

F. POWER ALLOCATION

The more power of pilots transmitted, the more accurate estimation of the channel is achieved in the proposed superimposed system. On the contrary, decrease of the power of information symbols causes performance degradation of symbol detection. The tradeoff problem of power allocation needs to be considered, which is beyond the scope of this paper, and a sub-optimal solution is proposed to maximize the signal to noise ratio (SNR) in [28]. We only illustrate the effect of the ratio of pilot and information power by numerical experiments in the next section.

From (13), average transmit power of a symbol in time domain on a single antenna is computed as:

$$\begin{aligned} E :&= \sum_{k=0}^{K-1} \frac{1}{PIK^2} \mathrm{tr}(\mathbf{X}(k)\mathbf{X}^H(k)) \\ &= \sum_{k=0}^{K-1} \frac{1}{PIK^2} \mathrm{tr}(\mathbf{S}_{\mathrm{c}}(k)\mathbf{U}(\mathbf{S}_{\mathrm{c}}(k)\mathbf{U})^H) \\ &\quad + \sum_{k=0}^{K-1} \frac{1}{PIK^2} \mathrm{tr}(\mathbf{P}_{\mathrm{c}}(k)\mathbf{V}(\mathbf{P}_{\mathrm{c}}(k)\mathbf{V})^H) \\ &:= E_{\mathrm{s}} + E_{\mathrm{p}}, \end{aligned} \quad (53)$$

where the former term is information power denoted as E_{s}, and the latter is pilot power denoted by E_{p}. Moreover, E_{s} is given by:

$$E_{\mathrm{s}} = \sum_{k=0}^{K-1} \frac{1}{PIK^2} \mathrm{tr}(||\mathbf{S}(k)||_F^2 \mathbf{I}_P) = \frac{N_{\mathrm{s}} B \sigma_s^2}{IK}. \quad (54)$$

The average transmit power of a pilot symbol is given by:

$$E_{\mathrm{p}} = \sum_{k=0}^{K-1} \frac{1}{PIK^2} \mathrm{tr}(\mathbf{P}_{\mathrm{c}}(k)\mathbf{P}_{\mathrm{c}}(k)^H). \quad (55)$$

The power of a total transmit symbol can be normalized by adapting the variance of information symbols σ_s^2 and the pilot power E_{p}, so that,

$$E = E_{\mathrm{p}} + E_{\mathrm{s}} = 1. \quad (56)$$

IV. NUMERICAL SIMULATIONS

In this section numerical experiments are performed to demonstrate the performance of the proposed integrated OFDM MIMO radar and communication system, and the EM algorithm based joint channel and information symbol estimation.

We first give the default experimental conditions. The element numbers of transmit and receiving antennas are $P = 2$, and $Q = 2$ respectively. The carrier frequency f_c of radar is 6GHz, the number of subcarrier K is 512, and the frequency spacing of subcarriers Δf is 0.195MHz, such that the bandwidth of the transmitted signal is 100MHz. The transmit power budget on every antenna is normalized to 1. The range cell number $N = 256$. The total number of OFDM symbols in a frame is $I = 4$. The number of consecutive OSTBC 4 codewords block is $B = 1$. The number of the original symbols per carrier is $N_{\mathrm{s}} = 2$, and the length of the codeword is $N_{\mathrm{c}} = 2$.

The number of the targets L is set to 30. Their amplitudes follow Gaussian distribution i.e., $a_{p,q,l} \sim N(0, \sigma_{\mathrm{t}}^2)$, where $\sigma_{\mathrm{t}}^2 = 1/L$ for normalization purpose of the channel. The time delays of the targets $\tau_{p,q,l}$ are uniformly distributed in N range cells. The Doppler frequencies of the targets follow Gaussian distribution i.e., $f_{\mathrm{D}p,q,l} \sim \mathcal{N}(0, \sigma_{\mathrm{D}}^2)$. The received SNR is defined as $SNR = 1/\sigma^2$. The proportion of pilot power equals to E_{p}, because the total transmit power is set to 1.

The Ambiguity function (AF) is usually used to evaluate the performance of radar waveforms, since it is equivalent to the pulse compression processing output. Here numerical analysis about the AF of the superimposed OFDM MIMO waveform is carried out to show the features which affect the radar signal processing. It is defined as the matched filtering 6 in the receiver:

$$\chi_p(\tau, f_{\mathrm{D}}) = \int_{-\infty}^{\infty} x_p(t) x_p^*(t+\tau) e^{j2\pi k f_{\mathrm{D}} t} dt, \quad (57)$$

where τ and f_{D} represent the time delay and Doppler frequency of the received signal, respectively. The OFDM signal has a range resolution of T/K and a Doppler resolution of $1/(T + T_{cp})$ [29].

The ambiguity diagram of the superimposed OFDM MIMO waveform is shown in Fig. 4. One can observe that there are two large sidelobes at time delays of $-T$ and T in the ambiguity diagram. That is because of the existence of the CP. The limit of radar range is restricted to the length of CP as mentioned earlier, so the ambiguity of time delay won't happen. The chirp-like knife edge can be clearly visualized by zooming into the relevant area as depicted in Fig. 5. In fact, the AF exhibits a thumbtack near the origin. This main thumbtack is larger than the others along the oblique line.

In Fig. 6, a channel estimation result (or so called range profile reconstruction for radar) for one pair of transmit and 8 receiving antennas is plotted with $SNR = 10$dB. We set $E_{\mathrm{p}} = 0.5$ by default. The real target amplitude is a dotted

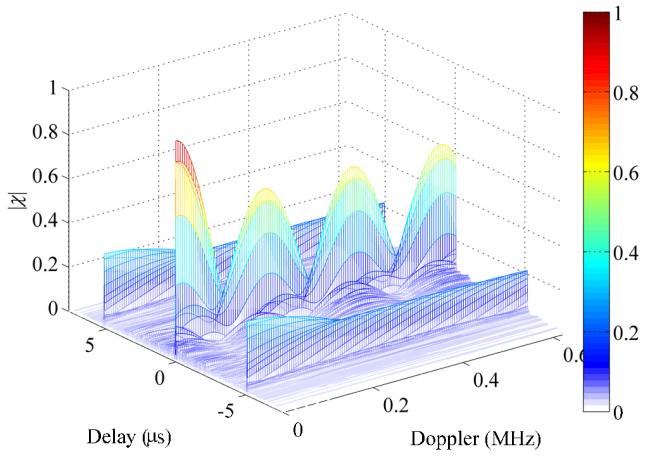


FIGURE 4. Ambiguity function of OFDM signal with ZC pilot.

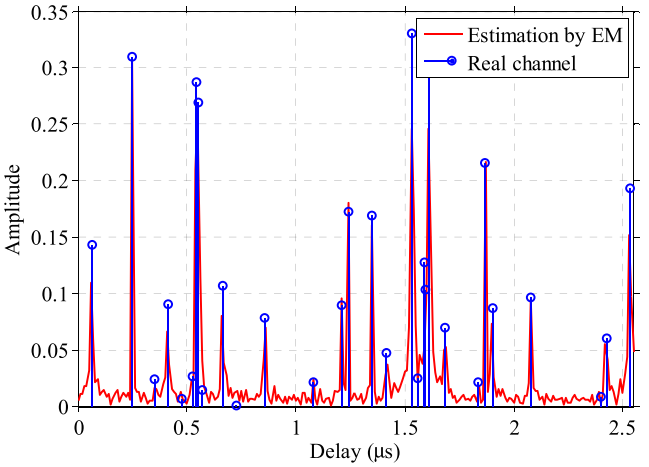


FIGURE 6. Channel estimation example with SNR = 10 dB.

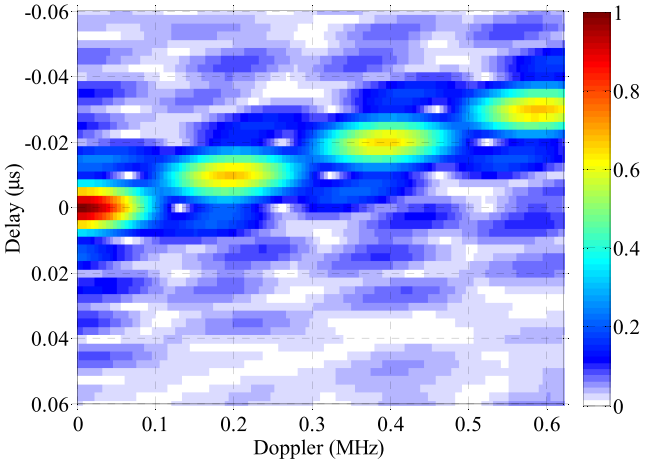


FIGURE 5. Zoomed in Ambiguity function of OFDM signal with ZC pilot.

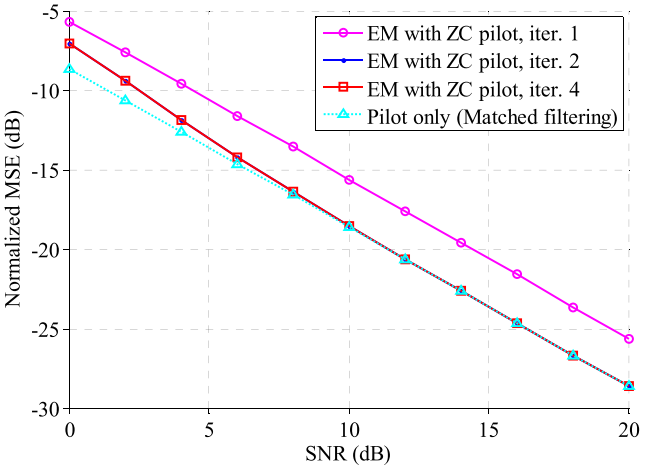


FIGURE 7. Normalized MSE of the channel versus SNR at different iterations.

line with circle marks, and the reconstruction result using the proposed EM algorithm is a solid line. The comparison shows that the proposed EM algorithm is precise in range reconstruction. In fact, contours of constant bistatic range are ellipses, with the transmitter and receiver as the two focal points. Targets lying on the ellipses with different bistatic ranges [30]. The MIMO radar has $P \times Q$ such range profiles. The direction of departure (DOD) and the direction of arrival (DOA) in every range bin can be calculated by the result of the range profiles. This topic is beyond the scope of this paper, please see more detail in [31], [32] for the joint estimation of DOA and DOD.

To evaluate the performance of the proposed EM algorithm, the normalized mean square error (MSE) of channel estimation versus SNR is plotted in Fig. 7. The normalized MSE is defined as $\text{NMSE} = ||\mathbf{H} - \hat{\mathbf{H}}||_F^2 \Big/ ||\mathbf{H}||_F^2$.

The solid line with circle marks is for the 1st iteration of the EM algorithm. It is the initialized estimation of the proposed algorithm shown in (39). In fact, the 1st iteration of the 10 channel estimation is equivalent to the algorithms discussed in [16], [17]. However, we improve the performance in an iterative way. The performances of the 2nd iteration and 4th iteration are demonstrated in the figure respectively by point and square marks. The two lines are very close to each other, which means that the EM algorithm using symbols estimation converges fast for a wide range of SNRs. The performance of channel estimation with no unknown information symbols is also plotted in a dotted line with triangle marks, served as a lower bound of the estimation and a benchmark. This is actually the matched filtering in frequency domain discussed in [3]. One can observe that the precision of the proposed EM algorithm using superimposed pilot performs almost the same compared with no unknown information symbols (matched filtering in frequency domain) with SNR>10dB. It can also carry unknown information at the same time. The symbol error rates (SER) of different iterations of the proposed algorithm are plotted in Fig. 8. From Fig. 7 and Fig. 8, it can be observed that the EM algorithm converges for just a few iterations (two to four iterations), especially at a moderate high SNR. The experiment shows

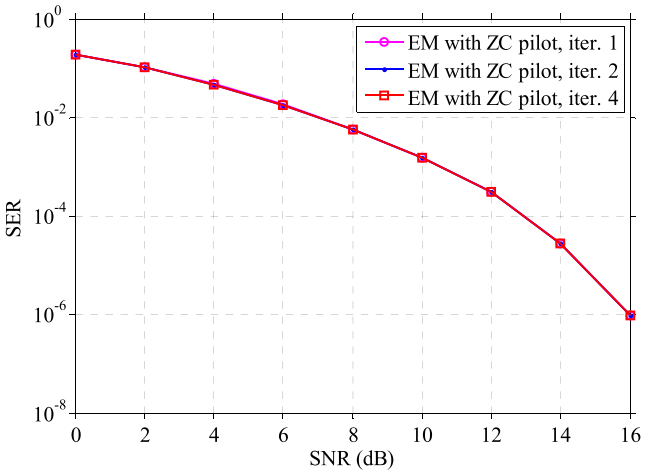


FIGURE 8. SER versus SNR at different iterations.

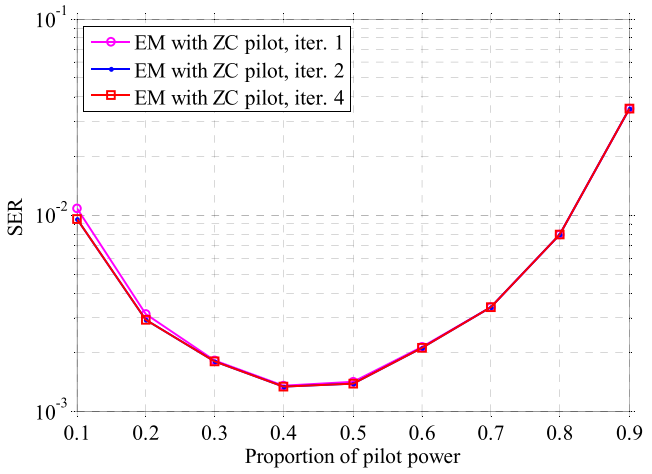


FIGURE 10. SER versus proportion of pilot power.

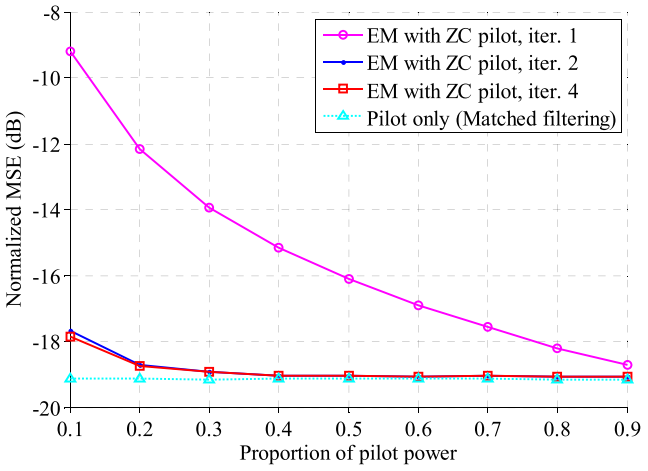


FIGURE 9. Normalized MSE versus proportion of pilot power.

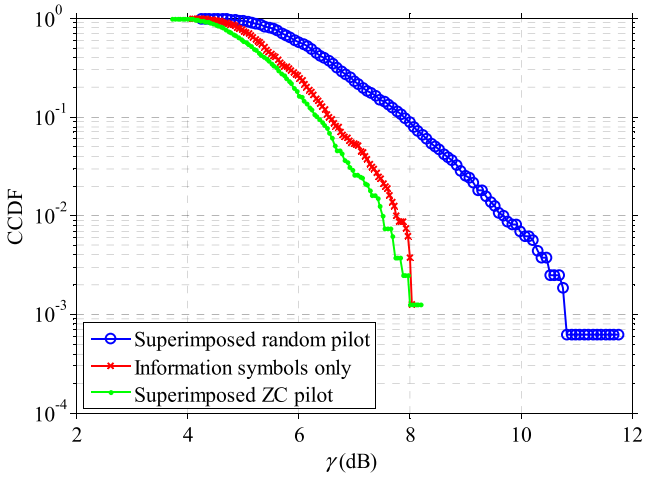


FIGURE 11. The PAPR comparison.

high convergence rate of the proposed EM algorithm, which implies one can get a fairly good performance by only two iterations in practical applications.

To evaluate the impact of the proportion of pilot power, E_{p}, in the total transmit power, experiments are carried out in Fig. 9 and Fig. 10, where SNR=10dB. From Fig. 9, we observe that normalized MSE monotonically decreases with the increase of the proportion of pilot power for the 1st iteration, which is natural because only pilots are contributed to the estimation of the channels. The curves after the 1st iteration are also monotonic, and coincident. The curves gradually converge to a constant value near $E_{\mathrm{p}} = 0.5$. The spacing between the curve for the 1st iteration and the others demonstrates that the information symbols also contribute to the target sensing. Fig. 10 shows the SER versus the proportion of the pilot power, E_{p}. One can observe that the lower the proportion of the pilot, the more improvement is achieved by the proposed algorithm. The SER performance improves obviously at the 2nd iteration. However, from the SERs at the 4th iteration, we also notice that almost no improvement is achieved after 2 iterations. And the SER improvement is not obvious when $E_{\mathrm{p}} > 0.3$. It shows that the SER is nearly a convex function of E_{p}, and an optimal value lies where $0.4 < E_{\mathrm{p}} < 0.5$. The decline of the left half curve is due to the contribution of the channel estimation, even though the information power decreases gradually.

OFDM signal suffers from large PAPR in time domain. The PAPR of the signal on the pth transmit antenna can be defined as

$$PAPR_p(t) = \frac{\max\limits_{0\le t<T} |x_p(t)|^2}{E[|x_p(t)|^2]}. \tag{58}$$

Fig. 11 shows the complementary cumulative distribution function (CCDF) of PAPR for different types of pilots, which is defined as $CCDF(\gamma) = \Pr(PAPR_p > \gamma)$. The solid line with 'x' marks is the CCDF of information symbols without superimposed pilots existing, where tone-reservation (TR) scheme [33] is used to reduce its PAPR. 16

The solid line with dot marks is the CCDF of information symbols with superimposed ZC pilot. A random pilot is

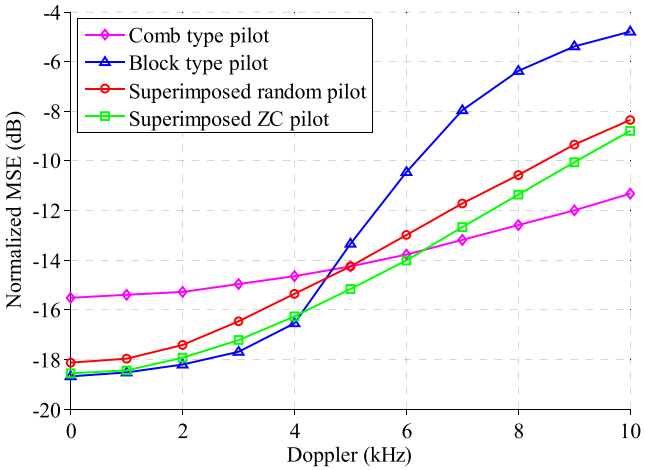


FIGURE 12. Normalized MSE of the channel versus Doppler frequency for different structures.

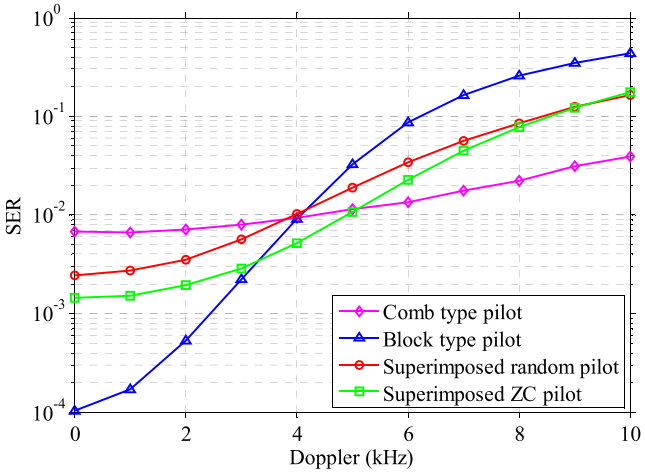


FIGURE 13. SER versus Doppler frequency for different structures.

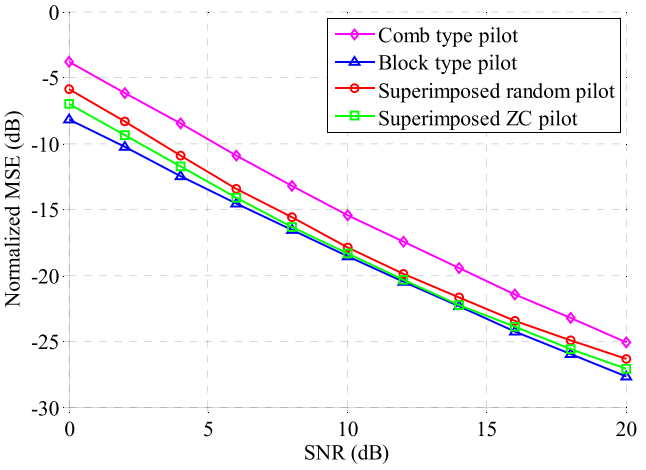


FIGURE 14. Normalized MSE of the channel versus SNR for different structures.

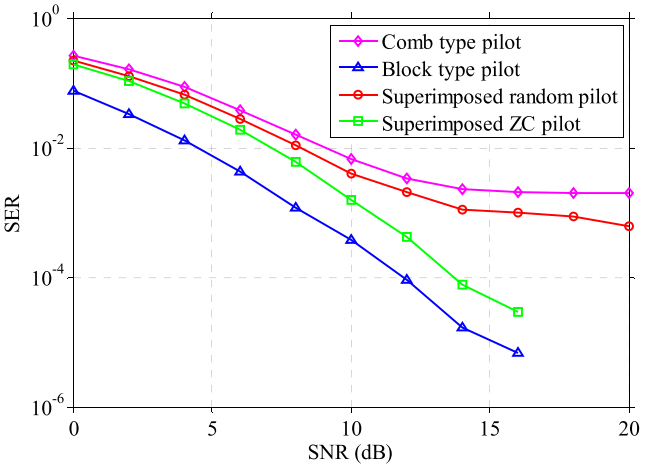


FIGURE 15. SER versus SNR for different structures.

alternatively used in (51) for comparison. The result is also plotted in Fig. 11 by a solid line with circle marks. Because of the constant amplitude property of the proposed ZC-based pilot, it slightly reduces the PAPR of the original information symbols, while, on the contrary, the simple random pilot increases the PAPR dramatically.

Consider the Doppler frequency tolerant performance of the superimposed OFDM system. The Doppler frequencies of targets can cause interchannel interference (ICI), such that they bring channel response and symbol estimation errors. Therefore, OSTBC based information estimation will be affected because we assume the channel keeps constant on CPI (I consecutive OFDM symbols). Therefore, the Doppler tolerance of the integrated system mainly depends on CPI. Next we will illustrate the Doppler effects by numerical experiments in the following section.

Fig. 12 and Fig. 13 are NMSE and SER respectively versus Doppler frequency standard deviation σ_{D} at SNR=10dB. Fig. 14 and Fig. 15 are NMSE and SER versus SNR at $\sigma_{\mathrm{D}} = 2$kHz. Green solid lines with square marks represent superimposed ZC pilot, and blue solid lines with triangle marks represent superimposed random pilot. Because the OSTC is used in the pilot, the waveform cross-correlation effects in MIMO system is mitigated [21]. It turns out that the proposed superimposed ZC pilot outweighs the simple random pilot as shown in the plots. The other types of pilot structures are also implemented in the simulation, including block type and comb type mentioned in Section II. For the sake of fairness, the average powers of the pilots for all the structures are equal to the powers of the information symbols respectively. For the block type, two successive pilot symbols are followed by two information symbols. For the comb type, half the subcarriers are pilot symbols. All the information and pilot symbols are OSTBC coded. The CPI of a superimposed pilot structure is the duration of four consecutive OFDM symbols. The CPI of the block type is the duration of four symbols, and that of the comb type is two. We observe that block type has the best performance in low Doppler frequency region, while the performance degrades rapidly when Doppler frequency grows. The comb type outweighs the others at high Doppler frequency, because

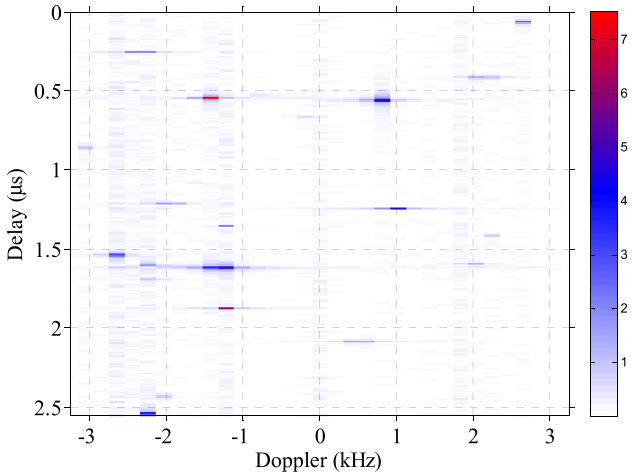


FIGURE 16. Range-Doppler spectrum with SNR=10dB.

of its shorter CPI. However, the proposed superimposed ZC pilot shows balanced and good performance under both the conditions. The simulation results are coincident with the aforementioned descriptions and the promise of the theoretical analysis.

The Doppler response of moving targets is also demonstrated in Fig. 16. The range-Doppler spectrum is calculated simply by taking the FFT of several consecutive estimated $\hat{\mathbf{H}}$ in slow-time domain, which is a traditional operation of radar, such as those discussed in [34], [35]. This experiment shows the ability of the superimposed OFDM MIMO radar in classical range-Doppler processing.

V. CONCLUSION

In this paper, a brand new OFDM MIMO integrated radar and communication system is proposed. The use of superimposed pilots with unknown information symbols as practical radar waveforms is addressed. The autocorrelation response of the waveform, i.e. typical response to point targets, and the Doppler processed response of moving targets in slow time are demonstrated.

Numerical experiments verify the promise that the superimposed pilot has good performance on both frequency selective and fast fading channels. This property of superimposed pilot is suitable in the integrated radar and communication system, especially when targets with large Doppler shift exist. Using the proposed EM algorithm with the expectation calculated from ML estimations of information symbols, the channel estimation shows almost as the same performance as classical matched filtering at a moderately high SNR.

APPENDIX

A. THE MATRIX $\mathbf{W}_U(k)$ OF THE NOISE

Similar to (32), $\mathbf{W}_{\mathrm{U}}(k) \in \mathbb{C}^{QN_{\mathrm{c}} \times B}$ has the same structure of $\mathbf{Y}_{\mathrm{U}}(k)$ as follows:

$$\begin{aligned}\mathbf{W}_{\mathrm{U}}(k) &:= \left[\mathrm{vec}(\mathbf{W}\mathbf{U}_0^H), \ldots, \mathrm{vec}(\mathbf{W}\mathbf{U}_{B-1}^H)\right] \\ &= \left[(\mathbf{U}_0^* \otimes \mathbf{I}_Q)\mathrm{vec}(\mathbf{W}), \ldots, (\mathbf{U}_{B-1}^* \otimes \mathbf{I}_Q)\mathrm{vec}(\mathbf{W})\right].\end{aligned} \tag{59}$$

From the definition of $\mathbf{W}_{\mathrm{U}}(k)$ in (59), one can get the expectation of the covariance matrix of $\mathbf{W}_{\mathrm{U}}(k)$

$$\begin{aligned}&\mathbb{E}(\mathbf{W}_{\mathrm{U}}(k)\mathbf{W}_{\mathrm{U}}^{\mathrm{H}}(k)) \\ &= \sum_{b=0}^{B-1} (\mathbf{U}_b^* \otimes \mathbf{I}_Q)\mathbb{E}[\mathrm{vec}(\mathbf{W})\mathrm{vec}(\mathbf{W})^H](\mathbf{U}_b^* \times \mathbf{I}_Q)^H \\ &= \sigma^2 \mathbf{I}_{QN_{\mathrm{c}}},\end{aligned} \tag{60}$$

where the unitary property of $\mathbf{U}$ is used in the second equation.

B. COVARIANCE MATRIX OF SYMBOL ESTIMATION ERRORS

Provided that the channel response is perfectly estimated, the detection error is merely determined by the noise $\mathbf{W}_{\mathrm{U}}(k)$. From (35), we get:

$$\mathbf{E}_{\mathrm{s}}(k) = \left(\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{A}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right) + j\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{B}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)\right) /||\hat{\mathbf{H}}_{\mathrm{f}}(k)||_F^2. \tag{61}$$

The mean value of the symbol estimation error $\mathbf{E}_{\mathrm{s}}(k)$ is:

$$\mathbb{E}(\mathbf{E}_{\mathrm{s}}(k)) = \mathbf{0}. \tag{62}$$

The covariance of the symbol estimation error $\mathbf{E}_{\mathrm{s}}(k)$ arising from the additive noise is given by:

$$\begin{aligned}\mathbb{E}(\mathbf{E}_{\mathrm{s}}(k)\mathbf{E}_{\mathrm{s}}^H(k)) = \mathbb{E}\Big(&\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{A}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)\ \mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{A}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)^T \\ &+ \mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{B}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{B}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)^T\Big) \\ &/||\hat{\mathbf{H}}_{\mathrm{f}}(k)||_F^4\end{aligned} \tag{63}$$

The first term can be expressed as

$$\begin{aligned}&\mathbb{E}\left(\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{A}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{A}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\right)^T\right) \\ &= \frac{1}{2}\mathbb{E}\left(\mathrm{Re}\left(\hat{\mathbf{G}}_{\mathrm{A}}^H(k)\mathbf{W}_{\mathrm{U}}(k)\mathbf{W}_{\mathrm{U}}^H(k)\hat{\mathbf{G}}_{\mathrm{A}}(k)\right)\right) \\ &= \frac{1}{2}||\mathbf{H}_{\mathrm{f}}(k)||_F^2\sigma^2\mathbf{I}_{N_{\mathrm{s}}},\end{aligned} \tag{64}$$

where the first equality follows from the identity that $\mathrm{Re}(\mathbf{X})\mathrm{Re}(\mathbf{Y}) = \frac{1}{2}\mathrm{Re}(\mathbf{XY} + \mathbf{XY}^*)$ for any two complex matrixes $\mathbf{X}$ and $\mathbf{Y}$, the former term $\mathbb{E}(\mathbf{XY})$ is zero since the covariance matrix of circularly symmetric complex Gaussian random vector is zero. The second equality follows from (60) and (36). Similarly, the second term of (63) can be derived, and (63) can finally be expressed as:

$$\mathbb{E}(\mathbf{E}_{\mathrm{s}}(k)\mathbf{E}_{\mathrm{s}}^H(k)) = \sigma^2/||\hat{\mathbf{H}}_{\mathrm{f}}(k)||_F^2\mathbf{I}_{N_{\mathrm{s}}}. \tag{65}$$

C. DERIVATIONS OF THE OPTIMAL SOLUTION

Notice that $p(\mathbf{S}|\mathbf{H})$ in (41) is independent of $\mathbf{H}$, which is an irrelevant term, and can be dropped. Because the noise in (29) is Gaussian distributed, $\mathcal{L}$ in (41) can be computed as:

$$\mathcal{L}(\mathbf{H}|\hat{\mathbf{H}}^{(\rho-1)}) = c - \frac{1}{\sigma^2}\sum_{m=0}^{RK-1}\left\|\mathbf{y}_{\mathrm{p}_m} - \mathbf{H}\mathbf{p}_m\right\|^2 - \frac{1}{\sigma^2}\sum_{m=0}^{(I-R)K-1}\mathbb{E}_{p(\mathbf{S}|\mathbf{Y}_\mathrm{p},\mathbf{Y}_\mathrm{d},\hat{\mathbf{H}}^{(\rho-1)})}\|\mathbf{y}_{\mathrm{d}_m} - \mathbf{H}\mathbf{s}_m\|^2, \tag{66}$$

where c is a constant, $\mathbf{y}_{\mathrm{p}_m}$, $\mathbf{p}_m$, $\mathbf{y}_{\mathrm{d}_m}$ and $\mathbf{s}_m$ are the mth columns of $\mathbf{Y}_\mathrm{p}$, $\mathbf{P}$, $\mathbf{Y}_\mathrm{d}$ and $\mathbf{S}$, respectively. The term σ^2 is the noise variance given in (29).

The LLF can be further written as:

$$\begin{aligned}&\mathcal{L}(\mathbf{H}|\hat{\mathbf{H}}^{(\rho-1)}):\\ &= c - \frac{1}{\sigma^2}\sum_{m=0}^{RK-1}\sum_{q=0}^{Q-1}\left\|Y_{\mathrm{p}_{q,m}} - \mathbf{h}_q\mathbf{p}_m\right\|^2 - \frac{1}{\sigma^2}\sum_{m=0}^{(I-R)K-1}\sum_{q=0}^{Q-1}\mathbb{E}_{p(\mathbf{S}|\mathbf{Y}_\mathrm{p},\mathbf{Y}_\mathrm{d},\hat{\mathbf{H}}^{(\rho-1)})}\left\|Y_{\mathrm{d}_{q,m}} - \mathbf{h}_q\mathbf{s}_m\right\|^2,\end{aligned} \tag{67}$$

where $Y_{\mathrm{p}_{q,m}}$ and $Y_{\mathrm{d}_{q,m}}$ are the (q,m)th entries of $\mathbf{Y}_\mathrm{p}$ and $\mathbf{Y}_\mathrm{d}$, respectively, and $\mathbf{h}_q \in \mathbb{C}^{PN\times 1}$ is the qth row of $\mathbf{H}$.

The likelihood function in (67) is a concave function of $\mathbf{h}_q$ [23]. Taking the complex gradient [36] of it with respect to $\mathbf{h}_q$, and setting the gradient to zero and solving the equation, one can obtain the maximum point at the ρth iteration as follows [23]:

$$\begin{aligned}&\mathbf{h}_q^{(\rho)^T}\\ &= \left(\sum_{m=0}^{RK-1}Y_{\mathrm{p}_{q,m}}\mathbf{p}_m^H + \sum_{m=0}^{(I-R)K-1}Y_{\mathrm{d}_{q,m}}\mathbb{E}\left(\mathbf{s}_m|\mathbf{Y}_\mathrm{p},\mathbf{Y}_\mathrm{d},\hat{\mathbf{H}}^{(\rho-1)}\right)^H\right)\\ &\quad\times\left(\mathbf{P}\mathbf{P}^H + \sum_{m=0}^{(I-R)K-1}\mathbb{E}\left(\mathbf{s}_m\mathbf{s}_m^H|\mathbf{Y}_\mathrm{p},\mathbf{Y}_\mathrm{d},\hat{\mathbf{H}}^{(\rho-1)}\right)\right)^{-1}.\end{aligned} \tag{68}$$

Stacking all the vectors $\mathbf{h}_q^{(\rho)}, q = 0,\ldots,Q-1$ of (68) together, we get a matrix form of the channel estimation for the ρth iteration as expressed in (43).

D. CALCULATION OF THE MEAN AND COVARIANCE

In the EM algorithm of [23], $\mathbf{S}$ is assumed to be non- informative Gaussian prior. The prior is only approximate, so the convergence of the procedure is slow. Here, we propose to employ the estimation of information symbols, $\hat{\mathbf{S}}$, to make full use of their OSTBC property.

By substituting (38) into (14), $\mathbf{S}_\mathrm{c}(k)$ can be expressed as its estimation value $\hat{\mathbf{S}}_\mathrm{c}(k)$ plus a noise $\mathbf{E}_\mathrm{c}(k)$

$$\begin{aligned}\mathbf{S}_\mathrm{c}(k) &= \mathbf{A}(\mathrm{Re}(\hat{\mathbf{S}}(k))\otimes\mathbf{I}_{N_\mathrm{c}}) + j\mathbf{B}(\mathrm{Im}(\hat{\mathbf{S}}(k))\otimes\mathbf{I}_{N_\mathrm{c}})\\ &\quad -\mathbf{A}(\mathrm{Re}(\mathbf{E}_\mathbf{S}(k))\otimes\mathbf{I}_{N_\mathrm{c}}) - j\mathbf{B}(\mathrm{Im}(\mathbf{E}_\mathbf{S}(k))\otimes\mathbf{I}_{N_\mathrm{c}})\\ &= \hat{\mathbf{S}}_\mathrm{c}(k) - \mathbf{E}_\mathrm{c}(k).\end{aligned} \tag{69}$$

where $\hat{\mathbf{S}}_\mathrm{c}(k) := \mathbf{A}(\mathrm{Re}(\hat{\mathbf{S}}(k))\otimes\mathbf{I}_{N_\mathrm{c}}) + j\mathbf{B}(\mathrm{Im}(\hat{\mathbf{S}}(k))\otimes\mathbf{I}_{N_\mathrm{c}})$, and $\mathbf{E}_\mathrm{c}(k) := \mathbf{A}(\mathrm{Re}(\mathbf{E}_\mathrm{s}(k))\otimes\mathbf{I}_{N_\mathrm{c}}) + j\mathbf{B}(\mathrm{Im}(\mathbf{E}_\mathrm{s}(k))\otimes\mathbf{I}_{N_\mathrm{c}})$ is an error term. Because $\mathbb{E}(\mathbf{E}_\mathrm{s}(k)) = \mathbf{0}$, we havečž

$$\mathbb{E}(\mathbf{E}_\mathrm{c}(k)) = \mathbf{0}. \tag{70}$$

Due to the property of OSTBC [18], the following equations hold:

$$\hat{\mathbf{S}}_\mathrm{c}(k)\hat{\mathbf{S}}_\mathrm{c}^H(k) = ||\hat{\mathbf{S}}(k)||_F^2\mathbf{I}_P, \tag{71}$$

$$\mathbf{E}_\mathrm{c}(k)\mathbf{E}_\mathrm{c}^H(k) = ||\mathbf{E}_\mathrm{s}(k)||_F^2\mathbf{I}_P = \mathrm{tr}(\mathbf{E}_\mathrm{s}(k)\mathbf{E}_\mathrm{s}^H(k))\mathbf{I}_P. \tag{72}$$

Substituting (65) into (72) and using (60) we get:

$$\mathbb{E}(\mathbf{E}_\mathrm{c}(k)\mathbf{E}_\mathrm{c}^H(k)) = N_\mathrm{s}\sigma^2\mathbf{I}_P. \tag{73}$$

Substituting $\mathbf{S}_\mathrm{c}(k) = \hat{\mathbf{S}}_\mathrm{c}(k) - \mathbf{E}_\mathrm{c}(k)$ into (28), we express $\mathbf{S}$ as its estimation $\hat{\mathbf{S}}$ plus an error $\mathbf{E}_\mathrm{d}$:

$$\mathbf{S} = \hat{\mathbf{S}} - \mathbf{E}_\mathrm{d}, \tag{74}$$

where

$$\hat{\mathbf{S}} = [(\mathbf{f}(0)\otimes\mathbf{I}_P)\hat{\mathbf{S}}_\mathrm{c}(0), ..., (\mathbf{f}(K-1)\otimes\mathbf{I}_P)\hat{\mathbf{S}}_\mathrm{c}(K-1)], \tag{75}$$

$$\mathbf{E}_\mathrm{d} = [(\mathbf{f}(0)\otimes\mathbf{I}_P)\mathbf{E}_\mathrm{c}(0), ..., (\mathbf{f}(K-1)\otimes\mathbf{I}_P)\mathbf{E}_\mathrm{c}(K-1)]. \tag{76}$$

Because the mean of $\mathbf{E}_\mathrm{d}$ is zero, the approximated value of (44) is given as in (46).

The covariance matrix of $\mathbf{S}$ can be computed as:

$$\begin{aligned}&\mathbb{E}\left(\mathbf{S}\mathbf{S}^H\right)\\ &= \mathbb{E}\left((\hat{\mathbf{S}}-\mathbf{E}_\mathrm{d})(\hat{\mathbf{S}}-\mathbf{E}_\mathrm{d})^H\right)\\ &= \mathbb{E}\left(\hat{\mathbf{S}}\hat{\mathbf{S}}^H\right) + \mathbb{E}\left(\mathbf{E}_\mathrm{d}\mathbf{E}_\mathrm{d}^H\right)\\ &= \sum_{k=0}^{K-1}(\mathbf{f}(k)\otimes\mathbf{I}_P)\hat{\mathbf{S}}_\mathrm{c}(k)\hat{\mathbf{S}}_\mathrm{c}^H(k)(\mathbf{f}(k)\otimes\mathbf{I}_P)^H\\ &\quad + \sum_{k=0}^{K-1}(\mathbf{f}(k)\otimes\mathbf{I}_P)\mathbb{E}\left(\mathbf{E}_\mathrm{c}(k)\mathbf{E}_\mathrm{c}^H(k)\right)(\mathbf{f}(k)\otimes\mathbf{I}_P)^H.\end{aligned} \tag{77}$$

Substituting (71) and (73) in (77), we get

$$\mathbb{E}\left(\mathbf{S}\mathbf{S}^H\right) = \sum_{k=0}^{K-1}||\hat{\mathbf{S}}(k)||_F^2\mathbf{I}_P + KN_\mathrm{s}\sigma^2\mathbf{I}_P. \tag{78}$$

Thus, the final covariance matrix of $\mathbf{S}$ can be expressed as in (47).

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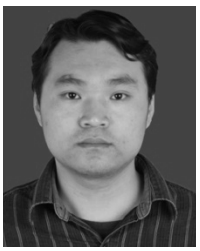
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